

Def. Let G be a group. Define

$$G^{(0)} = G \quad \text{and} \quad G^{(i+1)} = [G^{(i)}, G^{(i)}] = (G^{(i)})'$$

The descending chain

$$G = G^{(0)} \supseteq G^{(1)} \supseteq G^{(2)} \supseteq \dots \quad (i \in \mathbb{N})$$

is called the derived series of G .

Thm. Let G be a group. Then,

G is solvable if and only if $G^{(r)} = 1$ for some $r \in \mathbb{N} \cup \{0\}$.

Proof. Suppose that G is solvable. That is, there is a subnormal series $1 = H_r \triangleleft H_{r-1} \triangleleft \dots \triangleleft H_1 \triangleleft H_0 = G$ such that H_i/H_{i+1} is abelian for all $0 \leq i \leq r-1$.

Using induction for i , to show that $G^{(i)} \leq H_i$. If $i=1$, then

$$G' = G^{(1)} \leq H_1 \iff G/H_1 \text{ is abelian}$$

So, it is true by the assumption. Now, suppose that for $i=k$, $G^{(k)} \leq H_k$.

$$\text{Then, } G^{(k+1)} = [G^{(k)}, G^{(k)}] \leq [H_k, H_k]$$

and, since $H_{k+1} \triangleleft H_k$ with H_k/H_{k+1} is abelian, so $H_k' = [H_k, H_k] \leq H_{k+1}$, so $G^{(k+1)} \leq [H_k, H_k] \leq H_{k+1}$. So, by the induction, $G^{(r)} \leq H_r = 1$, so $G^{(r)} = 1$.

Conversely, Suppose that $G^{(r)} = 1$ for some $r \in \mathbb{N}$. (If $r=0$, then there are nothing to prove).

Then, the series $\{G^{(i)}\}_{i=0}^r$ is the normal series such that

$$G^{(i)} / G^{(i+1)} \text{ for all } 0 \leq i \leq r-1,$$

Since $G^{(i+1)}$ is the derived subgroup of $G^{(i)}$, so proved ~~QED~~

Prop. If G is solvable, then every subgroup and every quotient of G is solvable.

Proof. Let $r \in \mathbb{N}$ be a number such that $G^{(r)} = 1$.

Given a subgroup $H < G$, $H(i) \leq G(i)$ for all $0 \leq i \leq r$, so $H(r) \leq G^{(r)} = 1$.

Given a normal subgroup $N \triangleleft G$, observe that

$$(G/N)' = [G/N, G/N] = \langle [g_1N, g_2N] \mid g_1, g_2 \in G \rangle = \langle [g_1, g_2]N \mid g_1, g_2 \in G \rangle.$$

Meanwhile, since G' and N both are normal, $G'N$ is subgroup of G ,

moreover, $N \triangleleft G'N$, so $G'N/N$ is well-defined quotient group.

Given $[g_1, g_2] \cdot nN$, $[g_1, g_2] \cdot nN = [g_1, g_2]N \in \langle [g_1, g_2]N \mid g_1, g_2 \in G \rangle$,

and the reverse inclusion is given by every element in the generated subgroup is finite product of elements of $\{gN \mid g \in G'\}$, so

$$(G/N)' = G'N/N.$$

Inductively, $(G'N/N)' = (G'N)'N/N$